

Mohamad Latifian

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Research Interests

Computational Social Choice, Voting, Fair Division, Algorithmic Fairness, Mechanism Design, Algorithmic Game Theory

Education

Ph.D. in Computer Science

Sep. 2020 – Sep. 2024

University of Toronto, Canada

Advisors: Nisarg Shah and Allan Borodin

Thesis: Decoding Distortion: Pushing the Limits of Optimal Collective Decision-Making

GPA: 4+/4

M.Sc. in Algorithms and Computation

Sep. 2017 – Sep. 2019

Sharif University of Technology, Iran

Advisor: Mohammad Ghodsi

Thesis: Considering the Effect of Abstention in the Distortion Value.

GPA: 19.59/20

B.Sc. in Computer Engineering

Sep. 2012 – Jul. 2017

Sharif University of Technology, Iran

Advisor: Hamid Zarrabi-zadeh

Thesis: Hamiltonian Path in Solid Grid Graphs.

GPA: 17.22/20

Appointments

Postdoctoral Research Associate

Jul. 2024 – present

University of Edinburgh, UK

Host: Aris Filos-Ratsikas

Visiting Graduate Researcher

Sep. – Nov. 2023

University of Oxford, UK.

Host: Edith Elkind

Visiting Graduate Researcher

May – Jun. 2019

Max Planck Institute for Informatics, Germany.

Host: Alkmini Sgouritsa

Work title: Resource-Aware Protocols for Network Cost-Sharing Games (C3.)

Conference Publications

(Authors in alphabetical order)

- C15. M. Bullinger, E. Elkind, and M. Latifian. *Towards Fair and Efficient Public Transportation: A Bus Stop Model*. Proc. of the 24th International Conference on Autonomous Agents and Multi-Agent Systems (AAMAS), 2025 (Forthcoming).
- C14. E. Elkind, A. Lam, M. Latifian, T. Y. Neoh, and N. Teh. *Temporal fair division of indivisible items*. Proc. of the 24th International Conference on Autonomous Agents and Multi-Agent Systems (AAMAS), 2025 (Forthcoming).
- C13. S. Barman, S. Ebadian, M. Latifian, and N. Shah. *Fair Division with Market Values*. Proc. of the 39th AAAI Conference on Artificial Intelligence (AAAI), 2025 (Forthcoming).
- C12. M. Bedaywi, B. Flanigan, M. Latifian, and N. Shah. *Public Spirited Participatory Budgeting*. Proc. of the 39th AAAI Conference on Artificial Intelligence (AAAI), 2025 (Forthcoming).

- C11. M. Latifian and A. A. Voudouris. *The Distortion of Threshold Approval Matching*. Proc. of the 33rd International Joint Conference on Artificial Intelligence (IJCAI), pp. 2851–2859, 2024.
- C10. S. Ebadian, A. Filos-Ratsikas, M. Latifian, and N. Shah. *Computational Aspects of Distortion*. Proc. of the 23rd International Conference on Autonomous Agents and Multi-Agent Systems (AAMAS), pp. 1–14, 2023.
- C9. S. Ebadian, A. Filos-Ratsikas, M. Latifian, and N. Shah. *Explainable and Efficient Randomized Voting Rules*. Proc. of 37th Advances in Neural Information Processing Systems (NeurIPS), 36, 2024.
- C8. V. Gkatzelis, M. Latifian, and N. Shah. *Best of Both Distortion Worlds*. Proc. of the 24th ACM Conference on Economics and Computation (EC), pp. 738–758, 2023.
- C7. S. Ebadian, M. Latifian, and N. Shah. *The Distortion of Approval Voting with Runoff*. Proc. of the 22nd International Conference on Autonomous Agents and Multiagent Systems (AAMAS), pp. 1752–1760, 2023.
- C6. A. Kahng, M. Latifian, and N. Shah. *Voting with Preference Intensities*. Proc. of 37th AAAI Conference on Artificial Intelligence (AAAI), pp. 5697–5704, 2023.
- C5. A. Borodin, D. Halpern, M. Latifian, and N. Shah. *Distortion in Voting with Top- t Preferences*. Proc. of the 31st International Joint Conference on Artificial Intelligence (IJCAI), pp. 116–122, 2022.
- C4. A. Farhadi, M. Hajiaghayi, M. Latifian, M. Seddighin, and H. Yami. *Almost Envy-freeness, Envy-rank, and Nash Social Welfare Matchings*. Proc. of the 35th AAAI Conference on Artificial Intelligence (AAAI), pp. 5355–5362, 2021.
- C3. G. Christodoulou, V. Gkatzelis, M. Latifian, and A. Sgouritsa. *Resource-aware Protocols for Network Cost-sharing Games*. Proc. of the 21st ACM conference on economics and computation (EC), pp. 81–107, 2020.
- C2. M. Ghodsi, M. Latifian, and M. Seddighin. *On the Distortion Value of the Elections with Abstention*. Proc. of the 33rd AAAI Conference on Artificial Intelligence (AAAI), pp. 1981–1988, 2019.
- C1. M. Ghodsi, M. Latifian, A. Mohammadi, S. Moradian, and M. Seddighin. *Rent Division Among Groups*. Proc. of the 12th International Conference on Combinatorial Optimization and Applications (COCOA), pp. 577–591, 2018.

Journal Publications

- J1. M. Seddighin, M. Latifian, and M. Ghodsi. *On the Distortion Value of Elections with Abstention*. Journal of Artificial Intelligence Research, 70:pp. 567–595, 2021.

Working Papers

(Authors in alphabetical order)

- M3. A. Filos-Ratsikas, V. Gkatzelis, M. Latifian, E. Rewinski, and A. A. Voudouris. *Optimal Metric Distortion for Matching on the Line*. (Submitted to IJCAI)
- M2. M. Abbaszadeh, A. Ansarifard, M. Latifian, and M. Seddighin. *Metric Distortion with Preference Intensities*. (Submitted to IJCAI)
- M1. D. Afazeli, M. Latifian, and M. Seddighin. *Public Spirited Voting with a Network of Voters*.

Professional Service

- **PC Member:** EC '25, AAMAS '25, ICLR '25, AAAI '25, NeurIPS '24, IJCAI '24, AAMAS '24, AAAI '24, AAAI '21
- **Journal Reviewer:** Autonomous Agents and Multi-Agent System (JAAMAS), Artificial Intelligence (AIJ), Journal of Artificial Intelligence Research (JAIR)
- **Conference Reviewer:** STOC '25

Invited Talks

- T6. *Distortion in Social Choice Problems*, University of Edinburgh, EconCS Group Meetings (Oct. 2024).
- T5. *Fair Division with Market Values*, University of Edinburgh, Laboratory for Foundations of Computer Science (LFSCS) Lab Lunch (Oct. 2024).
- T4. *How to Make Efficient (Collective) Decisions*, Tehran Institute for Advanced Studies (Sep. 2024).
- T3. *Best of Both Distortion Worlds*, University of Edinburgh, Laboratory for Foundations of Computer Science (LFSCS) Seminar (Nov. 2023).
- T2. *Best of Both Distortion Worlds*, University of Essex, The Centre for Computational Finance and Economic Agents (CCFEA) Seminar Series (Oct. 2023)
- T1. *Resource-aware Protocols for Network Cost-sharing Games*, The Drexel EconCS Research Group Meetings (Jun. 2020).

Honors and Awards

- **C.C. Gotlieb (Kelly) Graduate Fellowship** in the Department of Computer Science, University of Toronto 2023
- **Silver medal** in the 21st Iranian National Olympiad in Informatics (INOI). 2011
- **Ranked 2nd and 7th** in Nationwide Graduate Entrance Exam, in Algorithms and Computation and Artificial Intelligence majors respectively. 2017
- **Ranked 1st** in cumulative GPA among 97 M.Sc. students of the department, Sharif University of Technology. 2019
- **Ranked 147th** out of 300,000 competitors in the national university entrance exam. 2012
- Scholarship from Iranian National Elites Foundation 2012 – 2019

Teaching Experience

Teaching Assistant

University of Toronto

- **Algorithm Design, Analysis & Complexity** Fall '23
- **Data Structures and Analysis** Winter '23
- **Algorithmic Game Theory and Mechanism Design** Fall '22

Teaching Assistant

Sharif University of Technology

(* := Lead TA)

- **Algorithmic Game Theory** Fall '17, Spring '19*
- **Design and Analysis of Algorithms** Fall '14, Fall '18*
- **Theory of Languages and Automata** Fall '14, Spring '15, Fall '15, Spring '16
- **Data Structures and Algorithms** Spring '15

Member of the Scientific Committee

*4th Sharif National Students Competition,
Sharif University of Technology.*

Fall 2016

Head of the Scientific Committee

*13th & 14th Mehregan Mathematic Workshop,
Shahid Ejeii High School.*

Aug. 2013 and Aug. 2014

Teaching Special Topics in Mathematics

*Teaching high-school students preparing for Olympiad in Informatics.
Topics included Algorithms, Graph Theory, Problem-Solving Strategies, and Programming.*

2011-2013

Supervision Experience

Mehrad Abbaszadeh and Ali Ansarifar (<i>Co-supervised with M. Seddighin</i>) Title: <i>Metric Distortion with Preference Intensities</i>	<i>Oct. 2023 – present</i>
Dana Afazeli Title: <i>Public Spirited Voting with a Network of Voters</i>	<i>Aug. 2024 – present</i>
Rayan Haddadi and Radin Jarireh Title: <i>Towards Optimal Ordinal MMS Approximation for Chores</i>	<i>Nov. 2024 – present</i>

Activities

- **Elected Member & President** of the Students Scientific Chapter (SSC)
Computer Engineering Department of Sharif University of Technology. 2014 – 2015
- **Chief of Staff** of 17th & 18th International Collegiate Programming Contest (ICPC)
Asia Region - Tehran Site. 2015 – 2016
- **Chief of Staff** of 6th & 7th AI Challenge 2014 – 2015
An event where AI agents written in Java, C++, or Python compete to pass the game field and overcome other agents.